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Vehicle cargo storage system

Abstract

A vehicle cargo storage system configured to be located in a floor of a vehicle is provided. The vehicle cargo storage system includes a chamber configured to extend into and below the floor of the vehicle, and a cargo holding structure configured to hold one or more cargo items and configured to move between a stowed position within the chamber and a deployed position outside the chamber. The vehicle cargo system also including a cover to a pivot mechanism to pivot the cover between a first closed position covering the one or more cargo items in the chamber and a second position to expose the one or more cargo items outside the chamber, and a locking mechanism configured to be actuated to lock the cover in the first closed position.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to vehicle storage systems, and more particularly relates to a vehicle having a convertible cargo storage system.

BACKGROUND OF THE DISCLOSURE

(2) Motor vehicles are commonly equipped with storage compartments for storing one or more

items. For example, the rear cargo area of a vehicle may include a hidden storage compartment that extends in a recessed area below the cargo floor and has a removable cover. Quite often, a spare tire or other items may be stored within the rear cargo area storage compartment. It would be desirable to provide for a cargo storage system that allows efficient access to the storage compartment.

SUMMARY OF THE DISCLOSURE

(3) According to a first aspect of the present disclosure, a vehicle cargo storage system configured to be located in a floor of a vehicle is provided. The vehicle cargo storage system includes a chamber configured to extend into and below the floor of the vehicle, a cargo holding structure configured to hold one or more cargo items and configured to move between a stowed position within the chamber and a deployed position outside the chamber, a cover connected to a pivot mechanism to pivot the cover between a first closed position covering the one or more cargo items in the chamber and a second position to expose the one or more cargo items outside the chamber, and a locking mechanism configured to be actuated to lock the cover in the first closed position.

(4) Embodiments of the first aspect of the present disclosure can include any one or a combination of the following features: the locking mechanism is hidden from view when the cover is in the first closed position; a remote input for actuating the locking mechanism between locked and unlocked states; the input comprises a key fob; the pivot mechanism comprises at least one pivot rod that rotates the cover at least approximately 180° between the first closed position and the second position; the at least one pivot rod is configured to extend into connectors in the floor of the vehicle; the cargo holding structure comprises one or more straps; the pivot mechanism comprises one or more hinges for rotating the cover about an edge; the cargo holding structure comprises a bin configured to hold the one or more cargo items; at least one control arm that pivots to allow the bin to move between the stowed and deployed positions; and the storage bin comprises a handle.

(5) According to a second aspect of the present disclosure, a vehicle is provided including a cargo floor defining a cargo area and a cargo storage system including a chamber configured to extend into and below the cargo floor, a cargo holding structure configured to hold one or more cargo items and configured to move between a stowed position within the chamber and a deployed position outside the chamber, a cover connected to a pivot mechanism to move between a first closed position covering the one or more cargo items within the chamber and a second position to expose the one or more cargo items outside the chamber, and a locking mechanism configured to be actuated to lock the cover in the first closed position.

(6) Embodiments of the second aspect of the present disclosure can include any one or a combination of the following features: the locking mechanism is hidden from view when the cover is in the first closed position; a remote input for actuating the locking mechanism between locked and unlocked states; the input comprises a key fob; the pivot mechanism comprises at least one pivot rod that rotates the cover at least approximately 180° between the first closed position and the second position; the at least one pivot rod is configured to extend into connectors in the cargo floor; the cargo holding structure comprises one or more straps; the pivot mechanism comprises one or more hinges for rotating the cover about an edge, and wherein the cargo holding structure comprises a bin configured to hold the one or more cargo items; and at least one control arm that pivots to allow the bin to move between the stowed and deployed positions.

(7) These and other features, advantages, and objects of the present disclosure will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

- (2) FIG. 1 is a rear perspective view of a motor vehicle having a cargo storage system shown in a stowed cargo position in a rear cargo area, according to a first embodiment;
- (3) FIG. 2 is a rear perspective view of the motor vehicle further illustrating the cargo storage system of FIG. 1 in a deployed cargo position;
- (4) FIG. 3A is an enlarged upper perspective view of the vehicle rear cargo area further illustrating the cargo storage system of FIG. 1 in the stowed cargo position;
- (5) FIG. 3B is a cross-sectional view taken through line IIIB-IIIB of FIG. 3A further illustrating the cargo storage system in the stowed cargo position;
- (6) FIG. 4A is an enlarged upper perspective view of the rear cargo area of FIG. 3A further showing the cargo storage system in the deployed cargo position;
- (7) FIG. 4B is a cross-sectional view taken through line IVB-IVB of FIG. 4A further illustrating the cargo storage system in the deployed cargo position;
- (8) FIG. 5 is an exploded view of the cargo storage system shown in FIG. 3A;
- (9) FIG. 6A is an enlarged upper perspective view of a vehicle rear cargo area equipped with a cargo storage system shown in the stowed cargo position, according to a second embodiment;
- (10) FIG. 6B is a cross-sectional view taken through line VIB-VIB of FIG. 6A further showing the cargo storage system in the stowed cargo position;
- (11) FIG. 7A is an enlarged upper perspective view of the vehicle rear cargo area having the cargo storage system shown in FIG. 6A in a deployed cargo position;
- (12) FIG. 7B is a cross-sectional view taken through line VIIB-VIIB of FIG. 7A further illustrating the cargo storage system in the deployed cargo position; and
- (13) FIG. 8 is an exploded view of the cargo storage system shown in FIG. 6A.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(14) Reference will now be made in detail to the present preferred embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts. In the drawings, the depicted structural elements are not to scale and certain components are enlarged relative to the other components for purposes of emphasis and understanding.

(15) As required, detailed embodiments of the present disclosure are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the disclosure that may be embodied in various and alternative forms. The figures are not necessarily to a detailed design; some schematics may be exaggerated or minimized to show function overview. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present disclosure.

(16) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the concepts as oriented in FIG. 1. However, it is to be understood that the concepts may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(17) The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to a vehicle and vehicle cargo storage system. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.

- (18) As used herein, the term “and/or,” when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items, can be employed. For example, if a composition is described as containing components A, B, and/or C, the composition can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.
- (19) In this document, relational terms, such as first and second, top and bottom, and the like, are used solely to distinguish one entity or action from another entity or action, without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.
- (20) As used herein, the term “about” means that amounts, sizes, formulations, parameters, and other quantities and characteristics are not and need not be exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art. When the term “about” is used in describing a value or an end-point of a range, the disclosure should be understood to include the specific value or end-point referred to. Whether or not a numerical value or end-point of a range in the specification recites “about,” the numerical value or end-point of a range is intended to include two embodiments: one modified by “about,” and one not modified by “about.” It will be further understood that the end-points of each of the ranges are significant both in relation to the other end-point, and independently of the other end-point.
- (21) The terms “substantial,” “substantially,” and variations thereof as used herein are intended to note that a described feature is equal or approximately equal to a value or description. For example, a “substantially planar” surface is intended to denote a surface that is planar or approximately planar. Moreover, “substantially” is intended to denote that two values are equal or approximately equal. In some embodiments, “substantially” may denote values within about 10% of each other, such as within about 5% of each other, or within about 2% of each other.
- (22) As used herein the terms “the,” “a,” or “an,” mean “at least one,” and should not be limited to “only one” unless explicitly indicated to the contrary. Thus, for example, reference to “a component” includes embodiments having two or more such components unless the context clearly indicates otherwise.
- (23) Referring to FIGS. 1 and 2, a wheeled automotive or motor vehicle **10** is generally illustrated configured with passenger seating for transporting one or more passengers, such as a driver of the vehicle and additional passengers, and for further transporting one or more items as cargo onboard the motor vehicle **10**. The motor vehicle **10** has a vehicle body **12** that generally defines a cabin interior **14**. The cabin interior **14** may contain various features and trim components. The cabin interior **14** is shown having an arrangement of passenger seats **16** which may include a front row of passenger seats and one or more rear rows of passenger seats, each configured to seat a passenger. The motor vehicle **10** shown herein, in one example, is a sport utility vehicle (SUV), however, it should be appreciated that the motor vehicle **10** may otherwise be configured as a van, a car or other motor vehicle having a cargo area configurable to include a cargo storage system.
- (24) The motor vehicle **10** is shown as a wheeled motor vehicle having a plurality of wheel assemblies **18** on a chassis which generally supports the body **12**. At the rear end of the motor vehicle **10** is a rear door liftgate **26** which is generally configured to which pivots about upper hinges between an open door position shown in FIG. 1 which exposes the cabin interior **14** via a rear opening **24** and a closed door position which closes the rear opening **24**. Located proximate to the rear of the motor vehicle **10** forward of the liftgate **26** and rearward of the passenger seats **16** is

a rear cargo area **20** which has a cargo floor **22**. One or more items of cargo may be located in the rear cargo area **20**.

(25) The motor vehicle **10** includes a cargo storage system **30** located in the rear cargo area **20**. The storage cargo system **30** is a convertible storage system that converts between a stowed cargo position shown in FIG. **1** and a deployed cargo position shown in FIG. **2**. In the stowed cargo position, the cargo is held in a chamber hidden below a lid or cover **32** on the cargo floor **22**. The cover **32** may be locked in a locked position with the cargo hidden in the stowed cargo position to prevent tampering or access to the cargo. In the deployed cargo position seen in FIG. **2**, the cargo is located above the cargo floor **22** and is exposed and accessible from the cargo area **20** near the rear of the motor vehicle **10**. The motor vehicle **10** may communicate with a remote device, such as a key fob **60**, which may actuate a lock mechanism to lock the cargo storage system **30** in the stowed cargo position to prevent access to the cargo and to unlock the lock mechanism when access to the cargo is allowed. It should be appreciated that other remote electronic devices such as a phone, an iPad, a computer, a remote switch onboard the vehicle, biometric inputs or voice or gesture interfacing communication input devices may be employed to actuate the lock mechanism between the locked and unlocked states.

(26) Referring to FIGS. **3A** and **3B**, the cargo storage system **30** is further shown in the stowed cargo position having the lid or cover **32** extending across an opening in the rear cargo floor **22** and covering a chamber **40** that extends into and below the cargo floor **22** of the vehicle **10**. The chamber **40** is formed as a recess within the rear cargo floor **22** and generally extends downward in a semi-cylindrical shape, according to one example. However, it should be appreciated that the chamber **40** may otherwise be configured in other shapes, such as a rectangle, for example. The chamber **40** may be formed separate from or integrally formed with the cargo floor **22**. The cover **32** is shown supported by and pivotally connected to the rear cargo floor **22** by a pivot mechanism that is configured to pivot the cover **32** between a first covered position as seen in FIGS. **3A** and **3B** which covers the cargo disposed within the chamber **40** and a second position as seen in FIGS. **4A** and **4B** that exposes the cargo outside and above the chamber **40**.

(27) The vehicle cargo storage system **30** includes a cargo holding structure shown as a storage bin **50** which is releasably connected via one or more straps **52** to strap connectors **54** on one side of the cover **32**. As such, the cargo holding structure is fixedly attached to one side of the cover **32**. The cover **32** has a pivot mechanism shown as a pair of pivot rods **34** that generally extend outward on opposite opposing sides and connect to the rear cargo floor **22**. The pivot rods **34** are generally disposed centrally within the cover **32** and allow the cover **32** to pivot along a central axis by at least 180° such that the cover **32** rotates between an upright first and upside-down second position. In the upright first position, the cover **32** has a strap handle **44** on the top side thereof which enables a user to engage the strap handle **44** to pivot the cover **32** between the first and second positions. The cover may pivot 180° from the stowed cargo position shown in FIGS. **3A** and **3B** to the deployed cargo position shown in FIGS. **4A** and **4B** when presented in the deployed position. The straps **52** may be unstrapped or otherwise disconnected to allow the bin **50** to be opened to expose one or more items of cargo.

(28) The pivot rods **34** are further shown extending into openings **36** on opposite sides of the cover **32** and further extending into openings **38** on a peripheral edge of the opening in the rear cargo door **32** as seen in FIG. **5**. As such, the pair of pivot rods **34** allows the cover **32** to rotate about a central axis extending through the pair of pivot pins **34**. It should be appreciated that the cover **32** may further be actuated with an actuator, such as electric motor **46** having an output drive shaft **48** configured to rotate at least one of the pivot pins **34**. As such, the cover **32** may be rotated manually by engaging the strap handle **44** or with the actuator via motor **46** actuated in response to an input such as from the key fob **60**.

(29) The vehicle cargo storage system **30** further includes a locking mechanism **42** which locks the cover **32** in the stowed cargo position to prevent access to the cargo within the chamber **40**. The

locking mechanism **42** may be configured as a pair of latches, in which at least one of the latches has a lock that may be actuated via remote control such as with a remote control device such as the key fob **60**. As such, access to the cargo stored within the chamber **40** may be controlled via the locking mechanism **42**.

(30) Referring to FIGS. **6A-8**, a vehicle cargo storage system **130** is illustrated, according to another embodiment. The vehicle cargo storage system **130** is shown generally formed in the rear cargo storage area within the rear cargo floor **22** which may be located on a motor vehicle **10** as shown in the embodiments of FIGS. **1** and **2**. In this embodiment, the vehicle cargo storage system **130** has a cover **132** hingedly connected at one edge with a pivot mechanism shown as a pair of hinges **142** to the rear cargo floor **22**, and has a handle with a strap **144** located on the opposite lateral edge. As such, a user may engage the handle strap **144** with force to pivot the cover **132** from a first closed cover position shown in FIGS. **6A** and **6B** to a second open cover position shown in FIGS. **7A** and **7B**.

(31) The vehicle storage system **130** has a chamber **140** formed extending into a recess below a lid or cover **132** of the rear cargo floor **122**. The chamber **140** may be integrally formed into the rear cargo floor **22** or may be formed separate therefrom. The cover **132** has a handle strap **144** and a pair of hinges **134** on a lateral edge. Disposed within the chamber **140** is a cargo holding structure that includes a bin **170** that moves between a stowed cargo position within the chamber **140** as seen in FIG. **6B** and an upward exposed cargo position extending above the chamber as shown in FIGS. **7A** and **7B**. The storage bin **170** is coupled to a pair of control arms **174** which pivot about a pair of pivot pins **176** on opposite lateral sides to allow the storage bin **170** to rotate approximately 90° as the bin **170** is moved from the stowed cargo position within the chamber **140** to the deployed cargo position outside and above the chamber **140**. The storage bin **170** has a handlebar **172** to enable a user to engage and pull or push on the storage bin **170** between the deployed and stowed positions. As such, a user may open the cover **132** and pull on the handlebar **172** of the storage bin to raise the storage bin **170** and rotate the storage bin **170** to a position where another bin **150** carrying cargo may be exposed and accessed by a user. It should be appreciated that the other bin **150** is shown having a container with a lid that may be opened to access the cargo stored therein.

(32) The vehicle cargo storage system **130** further includes a locking mechanism **142** which may be configured as a pair of latches, in which at least one latch has a lock. The locking mechanism **142** is hidden and allows for the cover **132** to be locked when the vehicle cargo storage system **130** is in the stowed cargo position, to thereby limit access to the cargo. The locking mechanisms **142** may be remotely actuated via one or more remote devices similar to those discussed above with respect to the first embodiment, such as with the use of a key fob or other remote communication device. As such, cargo items may remain locked within chamber **140** when desired to prevent access to the cargo.

(33) Accordingly, the vehicle cargo system **30** or **130** advantageously provides for the storage of cargo items onboard the motor vehicle **10**. The vehicle cargo system **30** or **130** is convertible between deployed and stowed positions and may advantageously provides for a lock mechanism to lock the cargo when the vehicle cargo system **30** or **130** is in the stowed cargo position.

(34) It is to be understood that variations and modifications can be made on the aforementioned structure without departing from the concepts of the present disclosure, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

Claims

1. A vehicle cargo storage system configured to be located in a floor of a vehicle, the vehicle cargo storage system comprising: a chamber configured to extend into and below the floor of the vehicle; a cargo holding structure configured to hold one or more cargo items and configured to move

between a stowed position within the chamber and a deployed position outside the chamber; a cover connected to a pivot mechanism to pivot the cover between a first closed position covering the one or more cargo items in the chamber and a second position to expose the one or more cargo items outside the chamber, wherein the pivot mechanism comprises at least one pivot rod configured to extend into connectors in the floor of the vehicle, and wherein the at least one pivot rod is generally disposed centrally within the cover to allow the cover to pivot along a central axis at least 180° to allow the cover to rotate between the first closed position and the second closed position; and a locking mechanism configured to be actuated to lock the cover in the first closed position.

2. The vehicle cargo storage system of claim 1, wherein the locking mechanism is hidden from view when the cover is in the first closed position.

3. The vehicle cargo storage system of claim 2 further comprising a remote input for actuating the locking mechanism between locked and unlocked states.

4. The vehicle cargo storage system of claim 3, wherein the input comprises a key fob.

5. The vehicle cargo storage system of claim 1, wherein the cargo holding structure comprises a storage bin and one or more straps.

6. A vehicle cargo storage system configured to be located in a floor of a vehicle, the vehicle cargo storage system comprising: a chamber configured to extend into and below the floor of the vehicle; a cargo holding structure comprising a bin configured to hold one or more cargo items and configured to move approximately 90° between a stowed position within the chamber and a deployed position outside the chamber; at least one control arm that pivots to allow the bin to move between the stowed and deployed positions; a cover connected to a pivot mechanism to pivot the cover between a first closed position covering the one or more cargo items in the chamber and a second position to expose the one or more cargo items outside the chamber, wherein the pivot mechanism comprises one or more hinges for rotating the cover about an edge; and a locking mechanism configured to be actuated to lock the cover in the first closed position.

7. The vehicle cargo storage system of claim 6, wherein the storage bin comprises a handle.

8. A vehicle comprising: a cargo floor defining a cargo area; and a cargo storage system comprising: a chamber configured to extend into and below the cargo floor; a cargo holding structure configured to hold one or more cargo items and configured to move between a stowed position within the chamber and a deployed position outside the chamber; a cover connected to a pivot mechanism to move between a first closed position covering the one or more cargo items within the chamber and a second position to expose the one or more cargo items outside the chamber, wherein the pivot mechanism comprises at least one pivot rod configured to extend into connectors in the floor of the vehicle, and wherein the at least one pivot rod is generally disposed centrally within the cover to allow the cover to pivot along a central axis at least 180° to allow the cover to rotate between the first closed position and the second closed position; and a locking mechanism configured to be actuated to lock the cover in the first closed position.

9. The vehicle of claim 8, wherein the locking mechanism is hidden from view when the cover is in the first closed position.

10. The vehicle of claim 9 further comprising a remote input for actuating the locking mechanism between locked and unlocked states.

11. The vehicle of claim 10, wherein the input comprises a key fob.

12. The vehicle of claim 8, wherein the cargo holding structure comprises one or more straps.

13. The vehicle of claim 8, wherein the pivot mechanism comprises one or more hinges for rotating the cover about an edge, and wherein the cargo holding structure comprises a bin configured to hold the one or more cargo items.

14. The vehicle of claim 13 further comprising at least one control arm that pivots to allow the bin to move between the stowed and deployed positions.
